SONGYIN WU

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EDUCATION

University of California, Santa Barbara
Ph.D. in Computer Science *Sep 2022 - Now*
GPA: 4.0 / 4.0

Peking University
B.S. in Computer Science (Turing Class, an elite program) *Sep 2017 - June 2021*
Major GPA: 89.45 / 100.00

PUBLICATIONS

Monocular Online Reconstruction with Enhanced Detail Preservation
***Songyin Wu**, Zhaoyang Lv, Yufeng Zhu, Duncan Frost, Zhengqin Li, Ling-Qi Yan, Carl Ren, Richard Newcombe, Zhao Dong*
ACM SIGGRAPH 2025 (Conference Track)

GFFE: G-buffer Free Frame Extrapolation for Low-latency Real-time Rendering
***Songyin Wu**, Deepak Vembar, Anton Sochenov, Selvakumar Panneer, Sungye Kim, Anton Kaplanyan, Ling-Qi Yan*
ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia 2024)

ExtraSS: A Framework for Joint Spatial Super Sampling and Frame Extrapolation
***Songyin Wu**, Sungye Kim, Zheng Zeng, Deepak Vembar, Sangeeta Jha, Anton Kaplanyan, Ling-Qi Yan*
ACM SIGGRAPH Asia 2023 (Conference Track)

Unified Gaussian Primitives for Scene Representation and Rendering
*Yang Zhou, **Songyin Wu**, Ling-Qi Yan*
Preprint

BSDF Importance Baking: A Lightweight Neural Solution to Importance Sampling General Parametric BSDFs
*Yaoyi Bai, **Songyin Wu**, Zheng Zeng, Beibei Wang, Ling-Qi Yan*
Preprint

RESEARCH EXPERIENCES

Meta Reality Labs Research
Research Scientist Intern *June 2024 - January 2025*

- Research project on interactive scene reconstruction.

Intel Corporation
Research Scientist Intern *June 2023 - September 2023*

- Research project on the generative model for general rendering tasks.

Microsoft Research Asia
Research Scientist Intern *December 2021 - July 2022*

- Research project on face re-animation and view synthesis.

Cornell Tech
Research Assistant Intern *June 2020 – May 2021*

- Research project on image-based relighting and blending.

TECHNICAL SKILLS

Programming: C/C++, Python
Software & Tool: PyTorch, OpencCV, OpenGL, Blender, Falcor, Unreal